

FMUSER FUQ-4459E Modulator 1536 IP Input - 48 QAM Output 3 GE Ports DVB-C Multiplexing Scrambling CA Headend TV Systems Product Overview

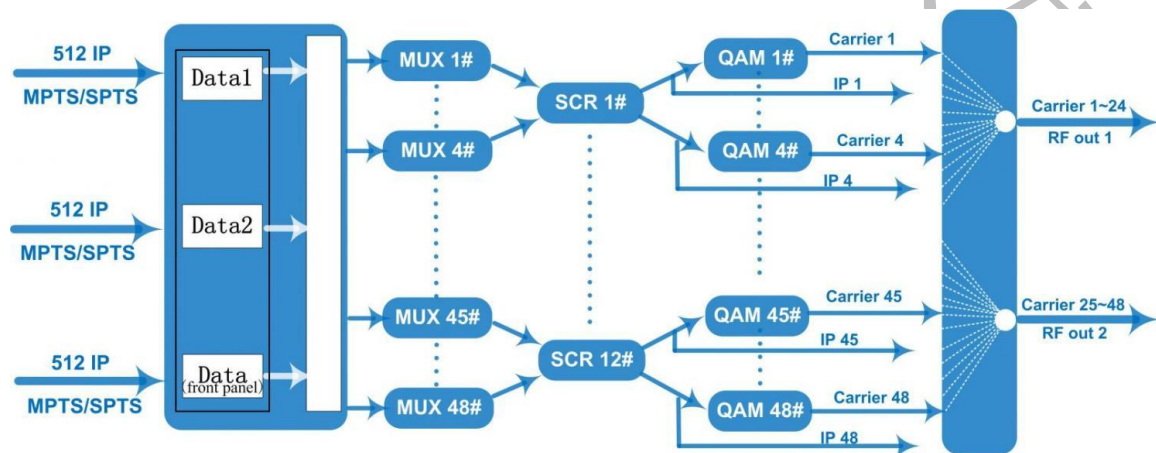


The FMUSER FUQ-4459E is a multiplexing, scrambling, and modulating device with max 1536 IP inputs via three GE ports and 48 QAM outputs through dual RF ports. It processes signals through multiplexing, scrambling, and modulation, including PID remapping and RS (204,188) encoding. Featuring web-based management, this device is ideal for CATV headend systems and broadcast video distribution.

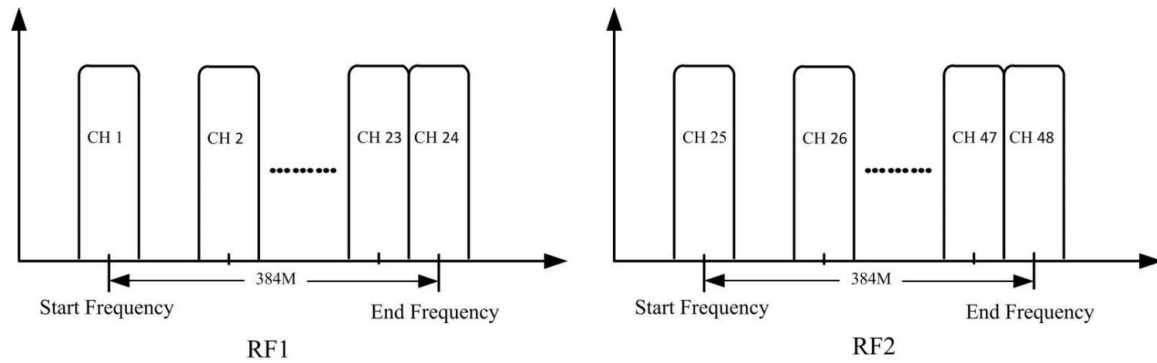
Key feature

- Handles max 1536 IP streams through three GE ports (SFP optional).
- Supports 6 simulcrypt CAs and 12 scrambling channels for security.
- Features PID remapping (up to 512 per channel), RS (204,188) encoding, and accurate PCR adjusting.
- Delivers 48 non-adjacent QAM carriers (50MHz~960MHz) and 48 IP outputs over UDP/RTP/RTSP.
- Provides web-based network management for configuration.

Inner Principle Chart



Carrier Setting Illustration



Specifications

Input	Input	Input	Max 1536 IP in thru 3 GE ports [Data 1 & Data 2 bi-directional ports, Data port (located on front panel), each port can input max 512 IP] 100/1000M self-adaption Ethernet Port (SFP interface optional)	Max 1536 IP in thru 3 GE ports [Data 1 & Data 2 bi-directional ports, Data port (located on front panel), each port can input max 512 IP] 100/1000M self-adaption Ethernet Port (SFP interface optional)
Input	Transport Protocol	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Transmission Rate	Transmission Rate	Max 840Mbps for each GE input	Max 840Mbps for each GE input
Mux	Input Channel	Input Channel	1536	1536
Mux	Output Channel	Output Channel	48	48
Mux	Max PIDs Remapping	Max PIDs Remapping	512 per output channel	512 per output channel
Mux	Functions	Functions	PID remapping (auto/manually optional)	PID remapping (auto/manually optional)
Mux	Functions	Functions	PCR accurate adjusting	PCR accurate adjusting

Mux	Functions	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling Parameters	Max simulcrypt CA	Max simulcrypt CA	6	6
Scrambling Parameters	Scramble Standard	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling Parameters	Scramble Channel	Scramble Channel	12	12
Scrambling Parameters	Connection	Connection	Local/remote connection	Local/remote connection
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83A	Constellation :16/32/64/128/256QAM
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83A	Bandwidth :8M
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83B/C	Constellation :64/256QAM
Modulation Parameters	DVB-C Modulator Section	DVB-C Modulator Section	J.83B/C	Bandwidth :6M
Modulation Parameters	QAM Channel	QAM Channel	48 non-adjacent carrier	48 non-adjacent carrier
Modulation Parameters	Modulation Standard	Modulation Standard	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C
Modulation Parameters	Symbol Rate	Symbol Rate	3600~7000Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM) 5361Ksps (J.83B, 256QAM)	3600~7000Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM) 5361Ksps (J.83B, 256QAM)
Modulation Parameters	Constellation	Constellation	16, 32, 64 , 128, 256QAM	16, 32, 64 , 128, 256QAM
Modulation Parameters	FEC	FEC	RS (204, 188)	RS (204, 188)

RF Output	Interface	Interface	2 F type output ports for 48 carriers, 75Ω Carrier 1~24 out thru RF1, 25~48 thru RF2	2 F type output ports for 48 carriers, 75Ω Carrier 1~24 out thru RF1, 25~48 thru RF2
RF Output	RF Range	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output	Output Level	Output Level	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping	-20dBm~+10dBm(87~117dbμV), 0.1dB stepping
RF Output	MER	MER	≥ 40dB	≥ 40dB
TS output	48 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M self-adaption, Ethernet Ports (SFP). IP 1~24 out thru Data1, IP 25~48 out thru Data2.	48 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M self-adaption, Ethernet Ports (SFP). IP 1~24 out thru Data1, IP 25~48 out thru Data2.	48 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M self-adaption, Ethernet Ports (SFP). IP 1~24 out thru Data1, IP 25~48 out thru Data2.	48 IP output over UDP/RTP/RTSP, unicast/multicast, 2*100/1000M self-adaption, Ethernet Ports (SFP). IP 1~24 out thru Data1, IP 25~48 out thru Data2.
System	Network management software (NMS) supporting	Network management software (NMS) supporting	Network management software (NMS) supporting	Network management software (NMS) supporting
General	Demission	420mm×440mm×44.5mm (W×L×H)	420mm×440mm×44.5mm (W×L×H)	420mm×440mm×44.5mm (W×L×H)
General	Weight	3kg	3kg	3kg
General	Temperature	0~45℃(operation), -20~80℃(storage)	0~45℃(operation), -20~80℃(storage)	0~45℃(operation), -20~80℃(storage)
General	Power Supply	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz or AC 220V±10%, 50/60Hz
General	Consumption	15.4W	15.4W	15.4W

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: +8613922702227

FMUSER 广州艾盛